

Active Directory Administration and Wazuh SOC Monitoring Lab

1. Executive Summary

This project demonstrates the deployment of a Windows Active Directory environment integrated with the Wazuh Security Information and Event Management (SIEM) platform.

The lab was built to simulate common enterprise administration and security monitoring activities. Active Directory was configured to manage users, groups, and policies, while Wazuh was deployed to collect and analyze security events generated throughout the environment.

The project validates the ability to detect, investigate, and document important Windows security events including failed logon attempts, user account creation, and privileged group membership changes.

2. Project Objectives

The objectives of this project were:

- Build a functional Active Directory environment.
- Configure Organizational Units (OUs).
- Create domain users and security groups.
- Deploy and manage Group Policy Objects (GPOs).
- Configure Windows security auditing.
- Deploy Wazuh agents to monitored endpoints.
- Investigate security events from a SOC analyst perspective.
- Produce portfolio-quality documentation.

3. Lab Environment

Infrastructure

Component	Purpose
Ubuntu Server	Wazuh Manager
Windows Server 2019	Domain Controller (RainhaStar)
Windows 11 Workstation	Domain Client
Additional Windows Workstation	Domain Client

Domain Information

Domain Name:

mydomain.com

Wazuh Version

4.14.5

Evidence

Insert Screenshot:

- VMware environment showing all virtual machines
- Windows Server 2019 dashboard
- Wazuh dashboard

Findings

The lab environment was successfully deployed using VMware Workstation. Active Directory services and Wazuh monitoring infrastructure were operational throughout the project.

Figure 1. VMware Workstation Lab Environment

(VMware Workstation environment showing all virtual machines used in the lab.)

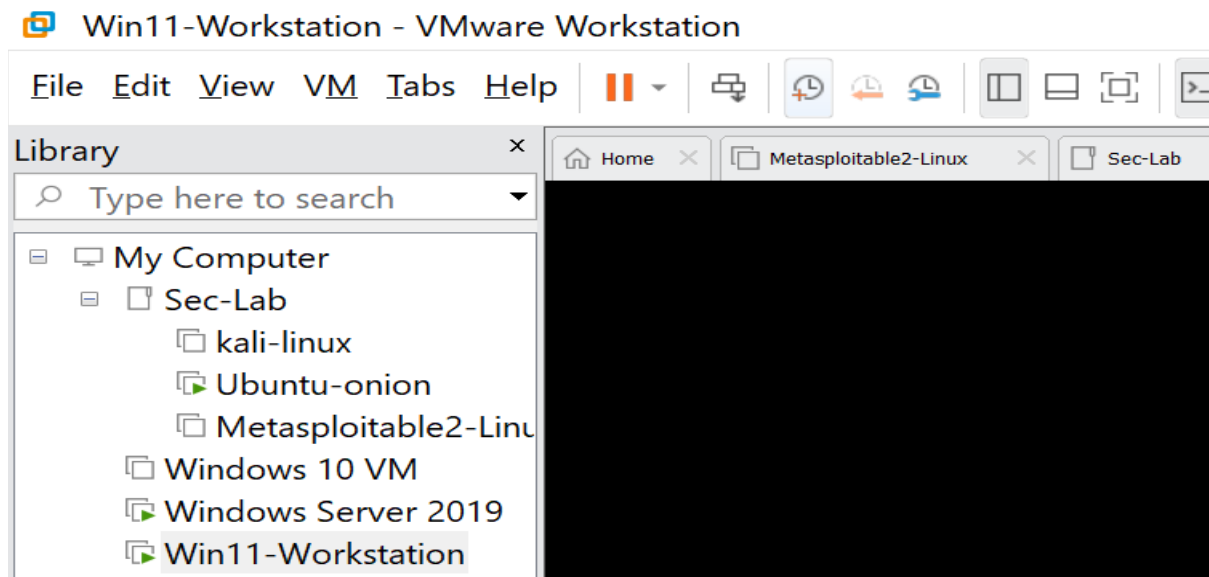


Figure 2. Windows Server 2019 Domain Controller

(Windows Server 2019 configured as the Active Directory Domain Controller.)

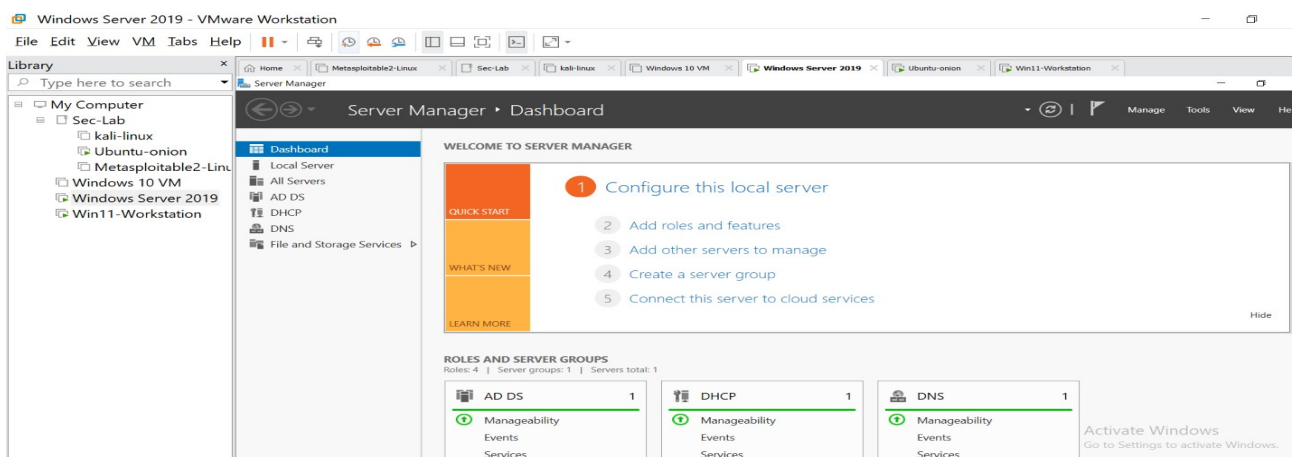
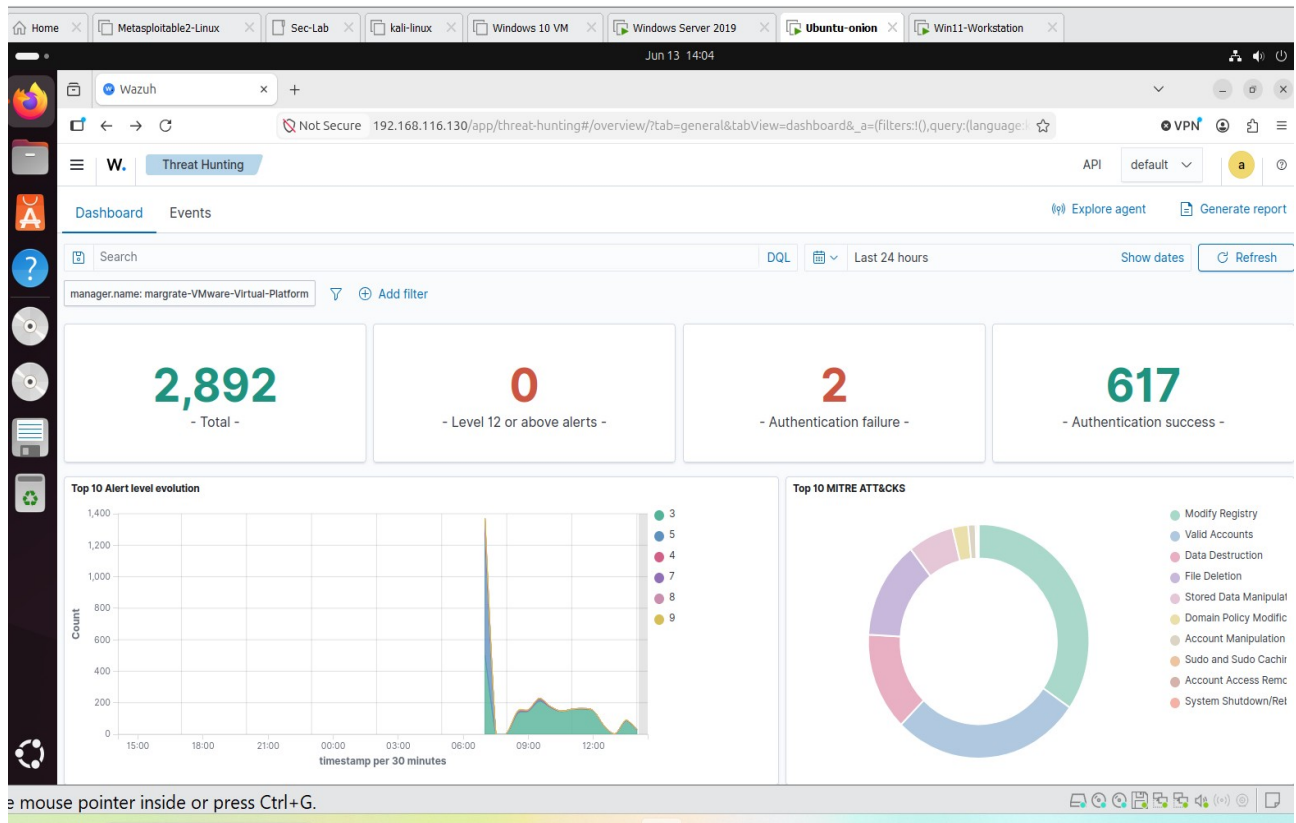


Figure 3. Wazuh SIEM Dashboard

(Wazuh dashboard used for centralized monitoring and analysis of security events.)



4. Active Directory Configuration

Objective

Configure Active Directory objects required for enterprise administration.

Activities Performed

- Created Organizational Units (OUs)
- Created security groups
- Created domain users
- Assigned users to groups
- Joined workstations to the domain

Findings

Active Directory was successfully configured on the Windows Server 2019 Domain Controller. Organizational Units (OUs), security groups, and user accounts were created and managed within the domain. Windows workstations were successfully joined to the mydomain.com domain, enabling centralized authentication and administration.

Figure 4. Organizational Unit Creation

(Creation of the IT Organizational Unit and verification that it appears in the Active Directory OU list.)

```
PS C:\Users\Administrator> New-ADOrganizationalUnit -Name "IT" -Path "DC=mydomain,DC=com"
PS C:\Users\Administrator> Get-ADOrganizationalUnit -Filter * | Select Name

Name
----
Domain Controllers
mydomain
Administrators
Domain Users
IT
PS C:\Users\Administrator>
```

Figure 5. Organizational Unit Verification

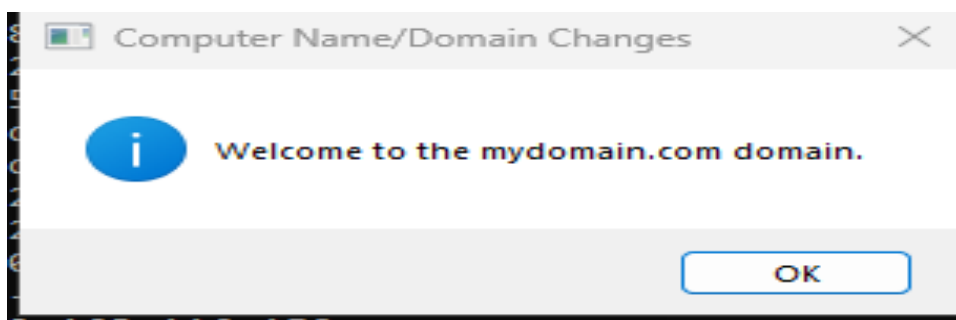
(Verification of the IT Organizational Unit using PowerShell.)

```
PS C:\Users\Administrator> Get-ADOrganizationalUnit -Identity "OU=IT,DC=mydomain,DC=com"

City           :
Country        :
DistinguishedName : OU=IT,DC=mydomain,DC=com
LinkedGroupPolicyObjects : {}
ManagedBy     :
Name           : IT
ObjectClass    : organizationalUnit
ObjectGUID     : 35d65e70-7944-4a20-bac8-0b97d5e66681
PostalCode     :
State          :
StreetAddress   :
PS C:\Users\Administrator>
```

Figure 6. Domain Join Confirmation

(Windows workstation successfully joined to the mydomain.com Active Directory domain.)



5. Group Policy Configuration

Objective

Configure Windows security auditing policies to generate security events for monitoring and analysis.

Activities Performed

- Created Workstation Security Policy GPO
- Configured Advanced Audit Policies
- Enabled Logon Auditing
- Enabled Account Management Auditing
- Linked GPO to Workstations OU
- Applied Group Policy updates

Evidence

Figure 7. Workstation Security Policy Configuration

(Group Policy Object created and configured to enforce Windows security auditing.)

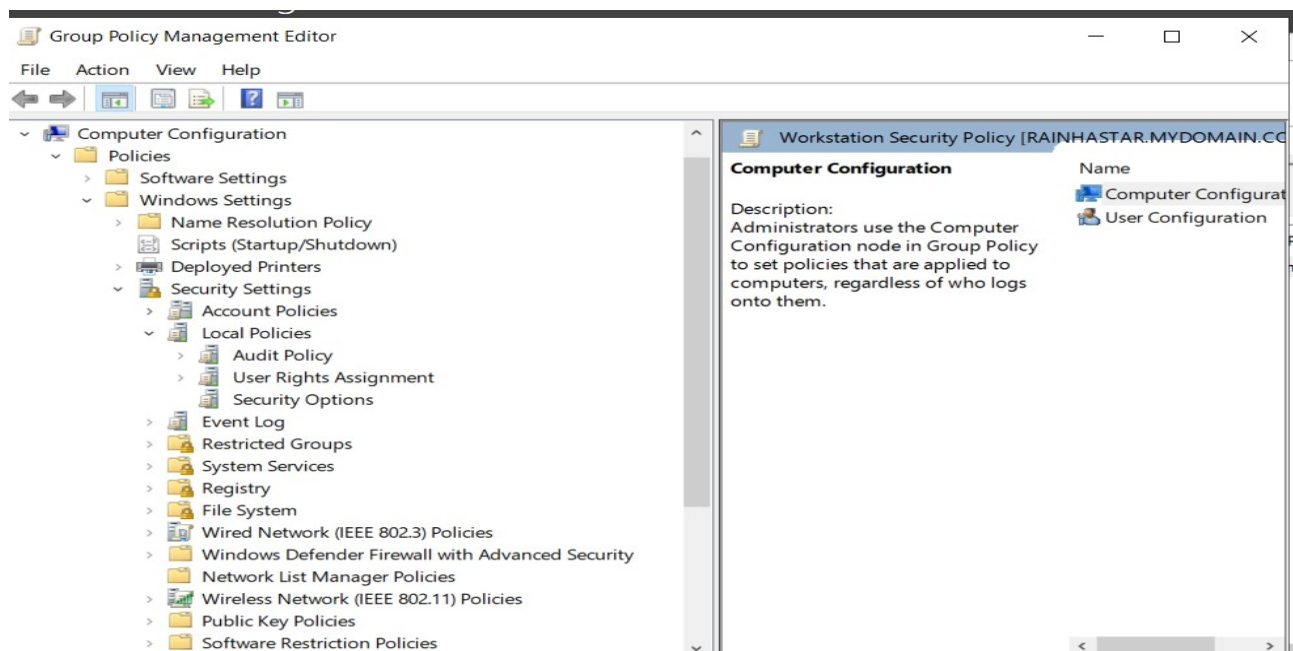


Figure 8. Group Policy Update Verification

(Successful application of Group Policy settings on the workstation.)

```
C:\Windows\System32>gpupdate /force
Updating policy...

Computer Policy update has completed successfully.
User Policy update has completed successfully.

C:\Windows\System32>
```

Findings

Windows security auditing was successfully configured through Group Policy. Audit settings were deployed to domain-joined workstations, enabling the generation of security events required for monitoring and investigation within Wazuh.

6. Event ID 4625 – Failed Logon Investigation

Objective

Generate a failed authentication event and validate its detection through Windows Security Logs and the Wazuh SIEM platform.

Procedure

1. Attempted to log in to a domain-joined workstation using an incorrect password.
2. Generated a failed authentication event.
3. Verified the event in the Windows Security Log.
4. Confirmed event collection and ingestion by the Wazuh agent.
5. Investigated the event details through the Wazuh dashboard.

Evidence

Figure 9. Failed Logon Attempt

(Failed authentication attempt using an incorrect password for the MYDOMAIN\jadmin account.)

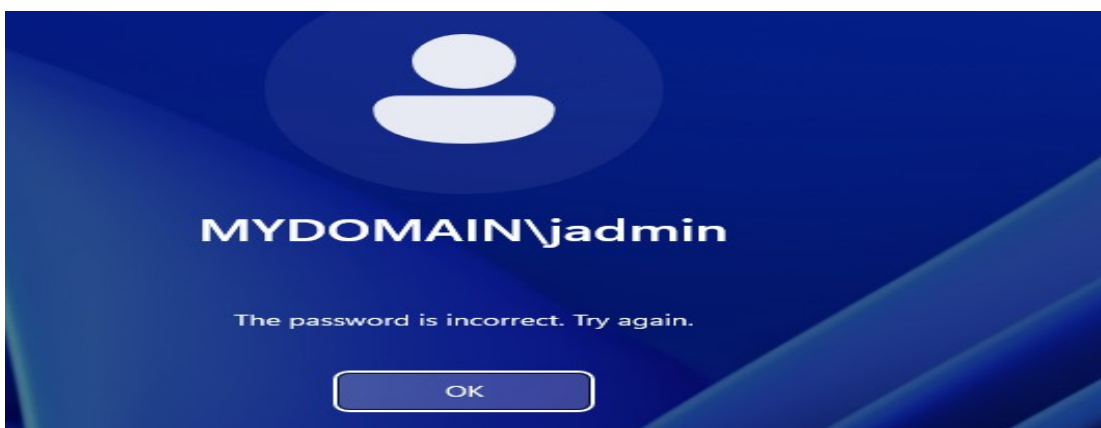


Figure 10. Event ID 4625 Verification

(PowerShell query confirming that failed authentication attempts were recorded as Event ID 4625 in the Windows Security Log.)

```
PS C:\WINDOWS\system32> Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4625} -MaxEvents 3 | Select TimeCreated, Id

TimeCreated      Id
-----
6/13/2026 7:29:22 AM 4625
6/13/2026 7:29:10 AM 4625
6/11/2026 9:03:50 PM 4625

PS C:\WINDOWS\system32>
```

Figure 11. Wazuh Detection of Event ID 4625
(Event ID 4625 successfully collected and displayed within the Wazuh SIEM platform.)

```
Process Name:\tC:\Windows\System32\consent.exe\r\n\r\nNetwork Information:\r\n\tWorkstation Name:\tDESKTOP-6FJEP7N\r\n\tSource Network Address:\t::1\r\n\tSource Port:\t\t0\r\n\r\nDetailed Authentication Information:\r\n\tLogon Process:\t\tCredPro\r\n\tAuthentication Package:\tNegotiate\r\n\tTransited Services:\t-\r\n\tPackage Name (NTLM only):\t-\r\n\tKey Length:\t\t0\r\n\r\nThis event is generated when a logon request fails. It is generated on the computer where access was attempted.\r\n\r\nThe Subject fields indicate the account on the local system which requested the logon. This is most commonly a service such as the Server service, or a local process such as Winlogon.exe or Services.exe.\r\n\r\nThe Logon Type field indicates the kind of logon that was requested. The most common types are 2 (interactive) and 3 (network).\r\n\r\nThe Process Information fields indicate which account and process on the system requested the logon.\r\n\r\nThe Network Information fields indicate where a remote logon request originated. Workstation name is not always available and may be left blank in some cases.\r\n\r\nThe authentication information fields provide detailed information about this specific logon request.\r\n\r\nTransited services indicate which intermediate services have participated in this logon request.\r\n\r\nPackage name indicates which sub-protocol was used among the protocols.\r\n\r\nKey length indicates the length of the generated session key. This will be 0 if no session key was requested.\r\n\r\n", "eventdata": {"subjectUserSid": "S-1-5-18", "subjectUserName": "DESKTOP-6FJEP7N$", "subjectDomainName": "MYDOMAIN", "subjectLogonId": "0x3e7", "targetUserSid": "S-1-0-0", "targetUserName": "administrator", "targetDomainName": "MYDOMAIN", "status": "0xc000006d", "failureReason": "%2313", "subStatus": "0xc000006a", "logonType": "2", "logonProcessName": "CredPro", "authenticationPackageName": "Negotiate", "workstationName": "DESKTOP-6FJEP7N", "keyLength": "0", "processId": "0x2e0", "processName": "C:\\Windows\\System32\\consent.exe", "ipAddress": "::1", "ipPort": "0"}}, "location": "EventChannel"}

margrate@margrate-VMware-Virtual-Platform:~$
```

Findings

A failed authentication attempt was generated by entering an incorrect password for a domain account. Windows recorded the activity as **Event ID 4625 (An account failed to log on)** in the Security log.

Analysis of the event showed that the authentication request was rejected due to invalid credentials. The event was successfully collected by the Wazuh agent and forwarded to the Wazuh Manager, where it was indexed and displayed for monitoring and investigation.

The investigation confirmed that:

- Windows Security Auditing was functioning correctly.
- The Wazuh agent successfully collected security logs.
- Security events were forwarded to the Wazuh Manager.
- Failed authentication attempts could be monitored and investigated through the SIEM platform.

No malicious activity was identified during the investigation. The event was intentionally generated for security monitoring validation and analyst training purposes.

Conclusion

The Active Directory Administration Lab was successfully implemented using Windows Server 2019, Windows 11, VMware Workstation, and Wazuh SIEM. Organizational Units, security groups, and domain users were created and managed through Active Directory. A workstation was successfully joined to the domain, and Group Policy Objects were configured to enforce Windows security auditing.

Security monitoring capabilities were validated by generating a failed logon event (Event ID 4625), which was successfully recorded within the Windows Security Log and detected by the Wazuh SIEM platform. The project demonstrated core enterprise administration tasks and provided hands-on experience with identity management, policy enforcement, and security event monitoring.

Skills Demonstrated

- Active Directory Administration
 - Organizational Unit (OU) Management
 - User and Group Management
 - Domain Administration
 - Group Policy Management
 - Windows Security Auditing
 - Event Log Analysis
 - Wazuh SIEM Monitoring
 - Security Event Investigation
 - Windows Server 2019 Administration
 - VMware Workstation Virtualization
 - Security Operations Center (SOC) Fundamentals
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